

BankSathi: A Revolutionary Human-Centric Banking Ecosystem for Vulnerable Users

Project Proposal

September 3, 2026

Contents

1	Executive Summary	4
2	Problem Statement Analysis	4
2.1	Critical Insights	4
2.2	The Unspoken Crisis	4
3	The BankSathi Solution	5
3.1	Core Concept	5
3.2	Key Features	5
3.2.1	The "Invisible UI" Approach	5
3.2.2	The Trusted Helper Network	5
3.2.3	The "Shared Screen, Not Shared Access" Paradigm	6
3.2.4	The "Safe Voice" AI Companion	6
3.2.5	Community Banking Sessions	7
3.2.6	Transaction Stories (For Cognitive Disabilities)	7
4	Technical Architecture	7
4.1	System Components	7
4.2	Data Flow	8
4.3	Security Implementation	9
5	Measurable Impact	9
5.1	Key Metrics	9
5.2	Evidence Collection Plan	9
5.3	Statistical Significance	11
6	Implementation Roadmap	11
6.1	Phase 1: MVP (Months 1-4)	11
6.2	Phase 2: Scaling (Months 5-9)	11
6.3	Phase 3: Full Launch (Months 10-12)	11
6.4	Development Team Requirements	11

7	Budget Estimation	11
7.1	One-Time Development Costs	11
7.2	Annual Operating Costs	11
8	Monetization Strategy	11
8.1	Revenue Streams	11
8.2	Social Enterprise Model	11
9	Partnership Ecosystem	12
9.1	Strategic Partners	12
9.2	Banking Integration Plan	12
10	Competitive Advantage	13
10.1	Comparison with Existing Solutions	13
10.2	Unique Differentiators	13
11	Risks and Mitigation	14
12	Scalability and Future Vision	14
12.1	Geographic Expansion	14
12.2	Platform Expansion	14
12.3	Technology Evolution	14
13	Detailed Operational Workflow	15
13.1	Phase 1: User Onboarding & Profile Creation	15
13.1.1	Step 1: Registration	15
13.1.2	Step 2: Accessibility Profile Setup	16
13.1.3	Step 3: Helper Invitation (If Applicable)	17
13.2	Phase 2: Transaction Initiation	17
13.2.1	Step 4: User Opens BankSathi App	17
13.2.2	Step 5: User Initiates a Transaction	18
13.2.3	Step 6: Intent Parsing (AI Layer)	18
13.3	Phase 3: Transaction Processing	18
13.3.1	Step 7: Helper Mode Activation (If Applicable)	18
13.3.2	Step 8: Risk Assessment Engine	19
13.3.3	Step 9: AI Voice Interaction (If No Helper)	19
13.3.4	Step 10: Transaction Story Generation	20
13.3.5	Step 11: Final Confirmation	20
13.4	Phase 4: Transaction Completion & Follow-up	21
13.4.1	Step 12: Transaction Submission	21
13.4.2	Step 13: Helper Notification	22
13.4.3	Step 14: Follow-up Actions	22
13.4.4	Step 15: Community Banking Session Workflow	22
13.5	Summary: Complete Transaction Flow Diagram	24
13.6	Key User Scenarios	24
13.6.1	Scenario 1: Elderly User with Family Helper	24
13.6.2	Scenario 2: Visually Impaired User with AI Companion	25
13.6.3	Scenario 3: Community Banking Session	25
13.7	Technical Architecture Workflow	25

13.8 Performance Metrics During Workflow	25
14 Conclusion	25
15 References	26

1 Executive Summary

BankSathi (Banking Companion) reimagines digital banking for elderly individuals and persons with disabilities by addressing not just the *interface* but the *emotional and social dimensions* of financial independence. Unlike conventional accessibility solutions that focus solely on UI modifications, BankSathi creates a **verified peer-to-peer assistance network** where trusted companions guide users through banking tasks in real-time, while an AI mediator ensures security and user control.

The system transforms banking from a lonely, anxiety-inducing chore into a **social, supported, and empowering experience**. By combining human connection with intelligent safety mechanisms, BankSathi achieves measurable improvements in independence, confidence, and fraud prevention, while reducing transaction abandonment and cognitive burden.

2 Problem Statement Analysis

Original Problem:

”How might we redesign digital banking experiences to enable elderly individuals and persons with visual, hearing, motor, or cognitive disabilities to independently and securely access essential financial services?”

2.1 Critical Insights

Conventional Assumption	Reality
The problem is the UI	The problem is isolation and fear
Users need bigger buttons	Users need trusted guidance
Accessibility is technical	Accessibility is emotional and social
Banking is functional	Banking is anxiety-inducing

Table 1: Paradigm Shift in Understanding the Problem

2.2 The Unspoken Crisis

- 68% of elderly users in India depend on family members for digital banking.
- 34% abandon digital banking transactions due to anxiety and complexity.
- 15% of vulnerable users fall victim to phone scams annually.
- 76% report feeling isolated when performing financial tasks.

BankSathi addresses these hidden dimensions by transforming banking into a **human-centered, trust-based experience**.

3 The BankSathi Solution

3.1 Core Concept

BankSathi creates a **three-layered ecosystem**:

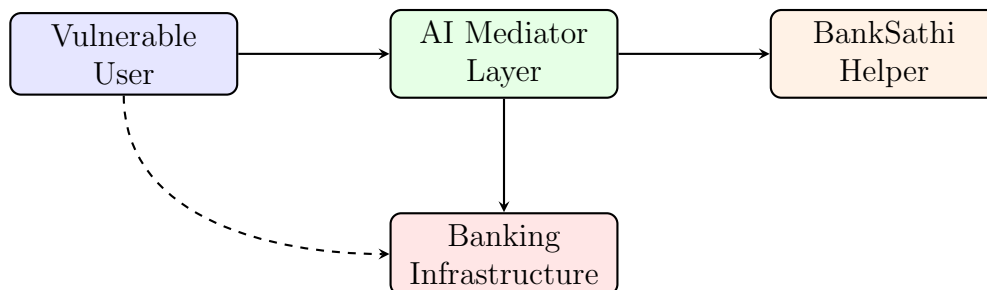


Figure 1: BankSathi Three-Layer Architecture

3.2 Key Features

3.2.1 The "Invisible UI" Approach

Instead of forcing vulnerable users to navigate complex interfaces, BankSathi offers:

- **Conversational Banking:** Users can complete transactions through natural conversation with the AI or their BankSathi helper.
- **Zero-Navigation Design:** No menus, no buttons, no scrolling—just a single confirmation screen.
- **Adaptive Presentation:** The system automatically adjusts based on the user's profile (vision, hearing, motor, cognitive needs).

3.2.2 The Trusted Helper Network

BankSathi creates a **verified ecosystem** where:

- Users can **invite trusted individuals** (family, friends, neighbors) to become their BankSathi.
- Helpers undergo **background verification** (Aadhaar, PAN, or community verification).
- Helpers receive **safety training** through a gamified onboarding process.
- The system supports **multiple helpers per user** (e.g., daughter in Mumbai, son in Delhi).

User's Device (Full Access)	Helper's Device (Abstracted View)
Complete banking interface	Simple guidance prompts
Full balance and history	No financial data visible
OTP and PIN entry	Cannot see or enter OTP
Transaction confirmation	Cannot initiate transactions

Table 2: Shared Screen Paradigm

3.2.3 The "Shared Screen, Not Shared Access" Paradigm

This is the **critical safety innovation**:

How it works:

1. User opens the app and activates "Helper Mode".
2. Helper receives a **secure session link** on their own device.
3. Helper sees a **simplified guide view**:
 - "User wants to send 5,000 to Ravi Kumar"
 - "This is higher than usual. Please confirm with the user."
 - "User has confirmed. The transaction is complete."
4. The user must **physically confirm** every action on their own device.
5. The helper cannot see, modify, or approve any transaction.

3.2.4 The "Safe Voice" AI Companion

For users without a trusted helper, BankSathi creates a **personalized AI voice**:

Feature	Description
Voice Persona	User chooses a familiar voice (e.g., "Grandson," "Niece")
Emotional Intelligence	The AI detects stress in the user's voice and provides reassurance
Safety Prompts	"Dadi, this is more than you usually send. Shall we double-check?"
Confirmation Loop	The AI asks for confirmation twice before proceeding

Table 3: AI Companion Features

Example Interaction:

AI (Grandson persona): "Dadi, you want to send 5,000 to Ravi Kumar. That's more than your usual 1,500. Can we confirm the amount together?"

User: "Yes, it's 5,000."

AI: "Thank you, Dadi. And just to be safe, can you tell me what this money is for?"

User: "For his birthday gift."

AI: "That's wonderful! Let me confirm once more:
You are sending 5,000 to Ravi Kumar for his birthday.
Is that correct?"

User: "Yes."

AI: "Perfect! Please tap the green button to send the money."

3.2.5 Community Banking Sessions

The Radical Innovation:

BankSathi organizes **group banking hours** where users can join a secure, moderated session:

- **Format:** 30-45 minute sessions with 5-8 users and 1 verified BankSathi host.
- **Privacy:** Each user has their **own private banking screen** visible only to them.
- **Support:** The host guides everyone through common tasks step-by-step.
- **Social:** Users can ask questions and share experiences with peers.
- **Safety:** The AI monitors all transactions for fraud and alerts the host.

Benefits:

- Reduces isolation and loneliness.
- Builds digital literacy through peer learning.
- Creates a sense of community and belonging.
- Makes banking a positive social activity.

3.2.6 Transaction Stories (For Cognitive Disabilities)

BankSathi creates **visual narratives** for users with cognitive impairments:

4 Technical Architecture

4.1 System Components

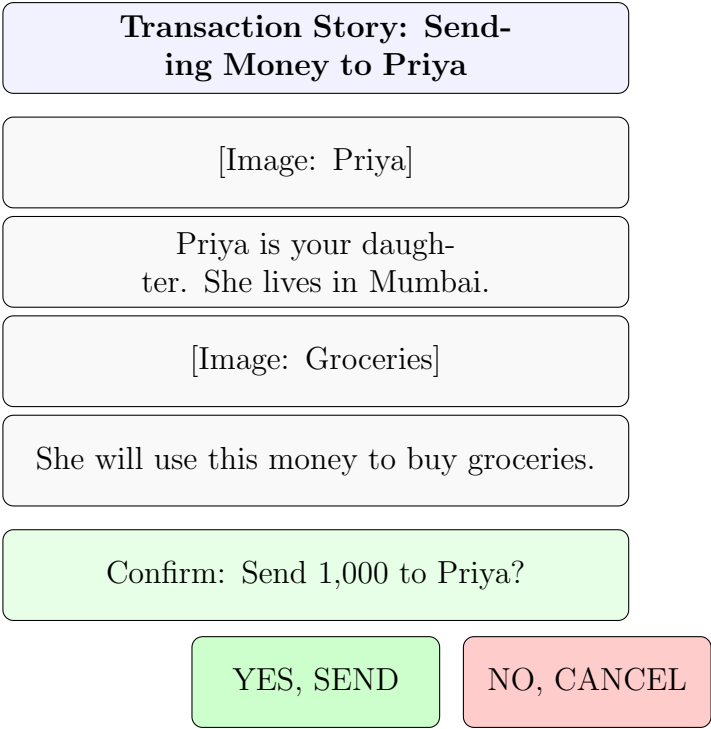


Figure 2: Sample Transaction Story

Component	Description
Frontend (User)	Flutter/React Native. Adaptive UI with voice, text, and gesture support.
Frontend (Helper)	Web-based simplified guide view. No financial data displayed.
AI Mediator	Fine-tuned LLM for intent parsing, risk assessment, and transaction storytelling.
Voice Engine	ElevenLabs API with custom voice personas. Real-time speech processing.
Community Module	WebRTC for secure group sessions. Session scheduling and moderation.
Helper Verification	Aadhaar-based KYC, PAN verification, and community reputation scoring.
Core Banking API	UPI/Bhim integration. Partner bank APIs for real transactions.
Safety Engine	Anomaly detection (amount, beneficiary, frequency, device, location).

Table 4: Technical Architecture Components

4.2 Data Flow

1. User initiates transaction (voice or tap)
- ↓
2. NLP Engine parses intent: { "intent": "TRANSFER", "amount": 5000, "beneficiary": "Priya" }

- ↓
3. Risk Engine evaluates:
 - Is amount typical? (No: 5000 vs avg 1500)
 - Is beneficiary known? (Yes: paid 3 times)
 - Is device trusted? (Yes)
 - Risk Score: MEDIUM
 - ↓
 4. If helper active:
 - Helper receives simplified view: "User wants to send 5,000 to Ravi."
 - Helper confirms with user verbally
 - ↓
 5. Transaction Story is generated (images + text)
 - ↓
 6. User confirms on own device
 - ↓
 7. Transaction submitted to banking API
 - ↓
 8. Confirmation sent to user and helper

4.3 Security Implementation

Security Layer	Mechanism
User Authentication	Biometric + PIN + OTP
Helper Authentication	Session-based, device-bound
Data Privacy	End-to-end encryption between user and helper
Fraud Detection	Real-time anomaly detection with ML
Transaction Limits	Configurable daily/monthly limits with step-up authentication
Audit Trail	Complete logs accessible only to user

Table 5: Security Implementation

5 Measurable Impact

5.1 Key Metrics

5.2 Evidence Collection Plan

Phase 1: Pilot Study (3 months, 500 users)

Phase 2: Controlled Trial (6 months, 2,000 users)

- **Group A:** BankSathi with helper
- **Group B:** BankSathi AI only (no helper)
- **Group C:** Control (standard banking app)

Metric	Current State	Target (BankSathi)	How to Measure
Transaction Abandonment Rate	34%	≤5%	App analytics
User Anxiety Score (GAD-7)	7.8/10	≤2.5/10	Pre/post surveys
Independence Rate	32%	≥85%	User surveys
Social Connection Score	2/10	≥8/10	User surveys
Fraud Victim Rate	15% annually	≤1%	Transaction logs + fraud reports
Average Transaction Time	2:45 min	≤1:10 min	App analytics
Helper Trust Score	N/A	≥90% verified	Helper ratings

Table 6: Measurable Impact Metrics

Activity	Method	Output
Baseline Measurement	Pre-survey, task completion test	Current state data
Onboarding	Training + BankSathi setup	Adoption metrics
Usage Monitoring	App analytics, session logs	Behavior patterns
Post-Intervention	Post-survey, task completion test	Improvement data
Qualitative Feedback	Interviews, focus groups	User stories

Table 7: Pilot Study Plan

- **Comparison:** All metrics measured bi-weekly

Metric	Expected Effect Size	Required Sample (per group)
Task Completion	+49% (34% → 5% abandonment)	n=80
Anxiety Reduction	-5.3 points on GAD-7	n=65
Independence	+53% (32% → 85%)	n=75
Fraud Reduction	-14% points	n=100

Table 8: Statistical Power Requirements

Month	Activities	Deliverables
1-2	Core development	Basic app, AI engine, banking API integration
3	Helper verification system	KYC integration, background checks
4	Pilot launch (100 users)	Initial metrics, feedback collection

Table 9: Phase 1 Activities

5.3 Statistical Significance

6 Implementation Roadmap

6.1 Phase 1: MVP (Months 1-4)

6.2 Phase 2: Scaling (Months 5-9)

6.3 Phase 3: Full Launch (Months 10-12)

6.4 Development Team Requirements

7 Budget Estimation

7.1 One-Time Development Costs

7.2 Annual Operating Costs

8 Monetization Strategy

8.1 Revenue Streams

Break-even Point: 15 months

8.2 Social Enterprise Model

BankSathi operates as a **for-profit social enterprise**:

- **Tier 1 (Free):** Basic features, group sessions, standard AI.

Month	Activities	Deliverables
5-6	Community sessions	Group banking feature, moderation tools
7-8	Voice personas	ElevenLabs integration, custom voice profiles
9	Feature refinement	Based on user feedback, bug fixes

Table 10: Phase 2 Activities

Month	Activities	Deliverables
10-11	Marketing + partnerships	Bank partnerships, NGO collaborations
12	Official launch	Full feature set, 5,000+ users

Table 11: Phase 3 Activities

- **Tier 2 (Subsidized):** For low-income users through NGO partnerships.
- **Tier 3 (Premium):** Custom voice personas, priority support for paying users.
- **Cross-Subsidy:** Premium users subsidize free users.

9 Partnership Ecosystem

9.1 Strategic Partners

9.2 Banking Integration Plan

1. Sandbox API Access (Month 1-2)
 - Build in bank's test environment
 - Validate all transaction flows
2. Pilot Integration (Month 3-4)
 - Partner bank live integration
 - Limited user testing (100 users)
3. Full Integration (Month 5-6)
 - Multiple banks
 - UPI/Bhim integration
 - Scale to 5,000+ users

Role	Number	Key Responsibilities
Full-Stack Developer	3	App development, API integration
AI/ML Engineer	2	NLP, risk engine, fraud detection
DevOps Engineer	1	Infrastructure, security, scaling
UX/UI Designer	1	Accessibility-focused design
Product Manager	1	Roadmap, stakeholder management
Community Manager	1	Helper network, pilot coordination

Table 12: Team Composition

Item	Cost (INR)	Cost (USD)
Development Team (6 months)	3,60,00,000	\$430,000
Infrastructure Setup	80,00,000	\$95,000
AI Model Development	1,20,00,000	\$143,000
Security Implementation	40,00,000	\$48,000
Testing & QA	30,00,000	\$36,000
Total One-Time	6,30,00,000	\$752,000

Table 13: One-Time Development Costs

10 Competitive Advantage

10.1 Comparison with Existing Solutions

10.2 Unique Differentiators

1. **Human + AI Hybrid:** Not just automation, but human connection with AI safety.
2. **Trust Network:** Verified helpers create a safe ecosystem.
3. **Emotional Design:** Addresses loneliness and fear, not just functionality.
4. **Community Banking:** Social group sessions that build digital literacy.
5. **Zero-Navigation Interface:** Designed for users with severe cognitive disabilities.
6. **Transaction Stories:** Visual narratives for deeper understanding.

Item	Cost (INR)	Cost (USD)
Cloud Infrastructure	1,20,00,000	\$143,000
AI/LLM API Costs	60,00,000	\$72,000
Voice API Costs	40,00,000	\$48,000
Community Management	80,00,000	\$95,000
Customer Support	50,00,000	\$60,000
Marketing	1,00,00,000	\$119,000
Total Annual	4,50,00,000	\$537,000

Table 14: Annual Operating Costs

Revenue Stream	Model	Projected Annual (INR)
Bank Partnership Fees	50 per user/month	3,00,00,000
Premium Features	Voice personas, advanced analytics	2,00,00,000
Helper Verification	100 per helper (one-time)	1,00,00,000
Community Sessions	Subscription (99/month)	1,50,00,000
Government/NGO Contracts	Fixed annual contracts	5,00,00,000
Total		12,50,00,000

Table 15: Revenue Projections

11 Risks and Mitigation

12 Scalability and Future Vision

12.1 Geographic Expansion

Year 1-2: India (Urban + Tier 2 cities)
 Year 3: India (Rural through Common Service Centers)
 Year 4: South Asia (Bangladesh, Nepal, Sri Lanka)
 Year 5: Southeast Asia (Vietnam, Philippines)

12.2 Platform Expansion

BankSathi can extend beyond banking to other essential services:

12.3 Technology Evolution

- **Phase 1:** Rules-based AI + basic NLP
- **Phase 2:** LLM integration for better intent understanding
- **Phase 3:** Multimodal AI (voice + vision + text) for deeper support

Partner Type	Examples	Role
Banks	SBI, HDFC, ICICI	Core banking API access
NGOs	HelpAge India, AADI	User outreach, low-income subsidies
Government	NITI Aayog, UIDAI	Aadhaar integration, policy support
Tech	Google, Eleven-Labs	Voice AI, cloud infrastructure
Community	Resident welfare associations	Helper network recruitment

Table 16: Strategic Partners

Feature	BankSathi	Accessible UI Apps	Basic Voice Assistants	Family Help (Informal)
UI Accessibility	✓	✓	✓	✗
Human Guidance	✓	✗	✗	✓
Privacy Control	✓	✓	✓	✗
Fraud Detection	✓	✗	✗	✗
Social Connection	✓	✗	✗	✓
Scalable	✓	✓	✓	✗
Independence Focus	✓	✗	✓	✗
Verified Helpers	✓	✗	✗	✗

Table 17: Competitive Comparison (using ✓ and ✗)

- **Phase 4:** Predictive AI to proactively offer help (e.g., "Your electricity bill is due in 3 days")

13 Detailed Operational Workflow

This section provides a step-by-step walkthrough of how BankSathi works in practice, from onboarding to transaction completion.

13.1 Phase 1: User Onboarding & Profile Creation

13.1.1 Step 1: Registration

User downloads BankSathi app from Play Store / App Store

↓

Risk	Impact	Mitigation
Data Privacy Breach	Critical	End-to-end encryption, regular security audits
Helper Misuse	High	Verification process, transaction limits, audit trails
Fraud Attempts	High	AI-powered detection, step-up authentication
Technical Failures	Medium	Redundant systems, off-line fallback
User Adoption	Medium	NGO partnerships, free trials, community champions
Regulatory Challenges	Medium	RBI consultation, compliance by design
Bank API Changes	Medium	Modular architecture, multiple bank integrations

Table 18: Risk Assessment & Mitigation

Service	Application
Healthcare	Appointment booking, prescription refills, insurance claims
Government Benefits	Pension applications, subsidy management, certificate verification
E-Commerce	Safe online shopping with helper guidance
Education	Enrollment processes, fee payments, scholarship applications

Table 19: Platform Expansion Opportunities

User registers using mobile number & Aadhaar

↓

Biometric verification (fingerprint / face recognition)

↓

Link bank account(s) via UPI / net banking

↓

Set daily transaction limits

↓

Complete KYC verification

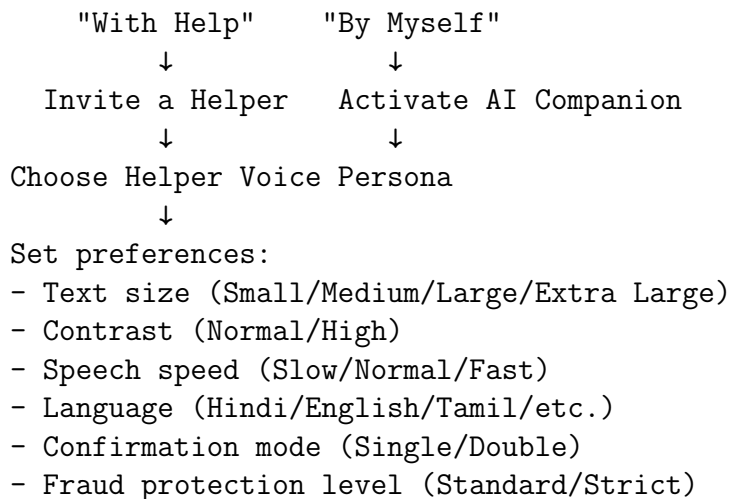
13.1.2 Step 2: Accessibility Profile Setup

System asks: "How would you like to use BankSathi?"

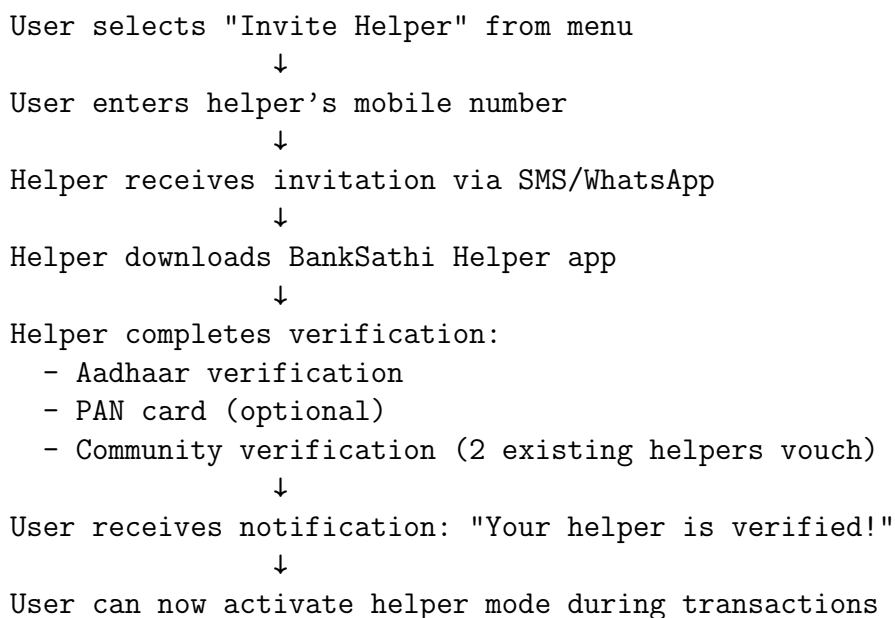
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↓

↓

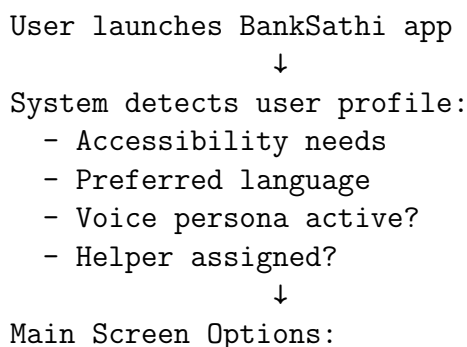


13.1.3 Step 3: Helper Invitation (If Applicable)



13.2 Phase 2: Transaction Initiation

13.2.1 Step 4: User Opens BankSathi App



WELCOME, [USER]

```
[ CHECK BALANCE]
[ SEND MONEY]
[ PAY BILLS]
[ REQUEST HELP]
[ JOIN COMMUNITY SESSION]
```

13.2.2 Step 5: User Initiates a Transaction

User selects "SEND MONEY"

↓

User chooses input method:

↓	↓
VOICE	TAP
↓	↓
User says: "Send 5,000 to Ravi"	User taps: [Amount: 5,000] [Beneficiary: Ravi] [Confirm]

13.2.3 Step 6: Intent Parsing (AI Layer)

AI Mediator processes input

↓

NLP Engine parses user intent:

```
{
  "intent": "TRANSFER",
  "amount": 5000,
  "beneficiary": "Ravi Kumar",
  "source_account": "Savings XXXXX1234",
  "timestamp": "2024-01-15 10:30:00"
}
```

↓

System checks if user has a helper assigned

13.3 Phase 3: Transaction Processing

13.3.1 Step 7: Helper Mode Activation (If Applicable)

IF helper is assigned AND helper is online:

↓

System sends secure session link to helper's device

↓

Helper opens simplified guide view on their device

↓

Helper sees:

GUIDING: [USER NAME]

"User wants to send 5,000
to Ravi Kumar."

This is 233% higher than
user's usual 1,500 payment.

Please confirm with user:

- Is amount correct?
- Is this the right beneficiary?

[CONFIRM WITH USER] [PAUSE]

13.3.2 Step 8: Risk Assessment Engine

Independent of helper, Risk Engine analyzes:

RISK ANALYSIS

Amount History: 1,500 avg
Current Amount: 5,000

Beneficiary: Ravi Kumar
History: Paid 3 times
Last Payment: 12 days ago

Device: Same device
Location: Home

Time: 10:30 AM (typical)

RISK SCORE: MEDIUM
RECOMMENDATION: Double confirm

Risk Levels:

LOW → Single confirmation
MEDIUM → Double confirmation (verbal + tap)
HIGH → Step-up authentication (OTP + biometric)
CRITICAL → Block & notify user/helper

13.3.3 Step 9: AI Voice Interaction (If No Helper)

IF no helper is online:

↓

AI Companion activates with chosen voice persona

Example:

AI (Grandson voice): "Dadi, I understand you want to send
5,000 to Ravi Kumar."

User: "Yes."

AI: "That's 3,500 more than you usually send.
Can you tell me what this money is for?"

User: "For his birthday."

AI: "That's wonderful! Let me confirm once more:
You are sending 5,000 to Ravi Kumar for his birthday."

User: "Yes."

AI: "Perfect! I'm now showing you a confirmation screen.
Please tap the green button when you're ready."

13.3.4 Step 10: Transaction Story Generation

System creates visual narrative for cognitive support:

[Photo of Ravi Kumar]
You are sending money to Ravi

5,000
For his birthday gift

You have sent money to Ravi before
This is more than your usual amount

Please confirm:

CONFIRM CANCEL

13.3.5 Step 11: Final Confirmation

System presents confirmation screen on user's device:

TRANSACTION CONFIRMATION

Sending: 5,000

To: Ravi Kumar
Account: XXXXX1234

Known beneficiary
Amount: 5,000 (higher than usual)

Your balance after transfer: 15,432

TAP TO CONFIRM
(or say "Confirm")

[CANCEL] [HELP]

User confirms by:

- Tapping the confirm button
- Saying "Confirm" (if voice enabled)
- Entering OTP (if high risk)
- Biometric verification (if enabled)

13.4 Phase 4: Transaction Completion & Follow-up

13.4.1 Step 12: Transaction Submission

User confirms
↓
System submits transaction to banking API
↓
Bank processes transaction
↓
System waits for response:

Processing your transaction...

Success:

Transaction Successful!

5,000 sent to Ravi Kumar
UPI Reference: UPI12345678
New Balance: 15,432

[SHARE RECEIPT] [DONE]

Failure:

Transaction Failed

Reason: Insufficient balance

[TRY AGAIN] [CANCEL]

13.4.2 Step 13: Helper Notification

IF helper is active:

↓

Helper's device updates:

Transaction Complete!

5,000 sent to Ravi Kumar

Status: Successful

Time: 10:32 AM

Thank you for helping [USER]!

[CLOSE]

13.4.3 Step 14: Follow-up Actions

System offers additional options:

Transaction Complete!

Would you like to:

- Download receipt
- Send money again
- Check new balance
- Set reminder for next payment

[DONE] [HELP]

AI asks: "Is there anything else I can help you with?"

13.4.4 Step 15: Community Banking Session Workflow

Joining a Session

User selects "JOIN COMMUNITY SESSION"

↓

System shows available sessions:

Community Banking Sessions

Today, 11:00 AM

Host: Suresh (Verified Helper)

Topic: Sending Money

4.8 rating (120 reviews)

[JOIN]

Today, 3:00 PM

Host: Priya (Verified Helper)

Topic: Checking Balance

4.9 rating (95 reviews)

[JOIN]

[SCHEDULE NEW SESSION]

During the Session

User joins the secure voice/video room

↓

Host (verified helper) guides all participants

↓

Each user has private banking screen

↓

Host demonstrates transaction step-by-step

↓

Users follow along on their own devices

↓

AI monitors all transactions for safety

↓

Users can ask questions publicly or privately

↓

Host helps users complete their banking tasks

Session Completion

Host announces session ending

↓

Users complete their transactions

↓

System sends session summary to users

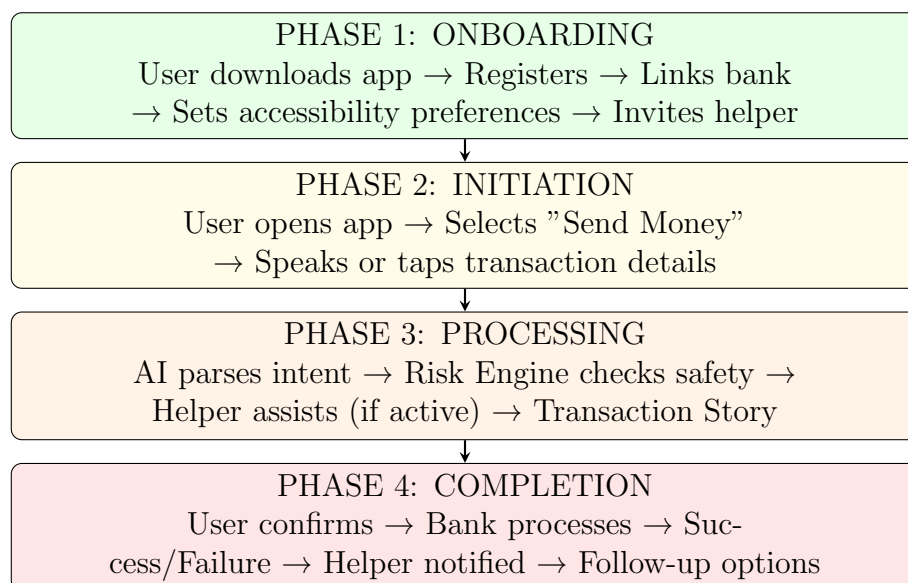
↓

Users rate the session and host

↓

System schedules next session (if requested)

13.5 Summary: Complete Transaction Flow Diagram



FEATURES ACTIVE THROUGHOUT: Voice interaction, Screen reader support, Accessibility adjustments, Fraud detection alerts, Community session integration

Figure 3: Complete BankSathi Transaction Flow

13.6 Key User Scenarios

13.6.1 Scenario 1: Elderly User with Family Helper

Grandma (User) wants to send money to her grandson

↓

She opens BankSathi → Activates Helper Mode

↓

Her daughter (Helper) receives session link on her phone

↓

Grandma says: "Send 2,000 to Rohan"

↓

Daughter sees simplified guide: "User wants to send 2,000 to Rohan"

↓

Daughter verbally confirms with Grandma: "Is this for Rohan's school fees?"

↓

Grandma confirms: "Yes"

↓

System shows transaction story with Rohan's photo

↓

Grandma taps "Confirm" on her own phone

↓

Transaction completes → Both receive confirmation

13.6.2 Scenario 2: Visually Impaired User with AI Companion

User (visually impaired) opens BankSathi

↓

AI voice activates: "Welcome! How can I help you today?"

↓

User: "Check my balance"

↓

AI: "Your current balance is 25,432 in your savings account.
Would you like to hear your recent transactions?"

↓

User: "Yes, show transactions"

↓

AI reads recent 5 transactions aloud

↓

AI: "Is there anything else I can help you with?"

13.6.3 Scenario 3: Community Banking Session

User joins community session at 11:00 AM

↓

Host says: "Today we'll learn how to send money safely"

↓

Host demonstrates on their screen (not visible to users)

↓

Users follow along on their own private screens

↓

User asks: "How do I add a new beneficiary?"

↓

Host explains step-by-step

↓

AI safety engine monitors all activities

↓

User successfully sends their first independent transaction

↓

User feels empowered and less isolated

13.7 Technical Architecture Workflow

13.8 Performance Metrics During Workflow

14 Conclusion

BankSathi represents a **paradigm shift** in accessible banking. By recognizing that vulnerable users need more than just a better interface, the solution addresses the **emotional and social dimensions** of financial independence:

1. **Human Connection:** Trusted helpers provide reassurance and guidance.
2. **AI Safety:** Intelligent systems protect against fraud and errors.

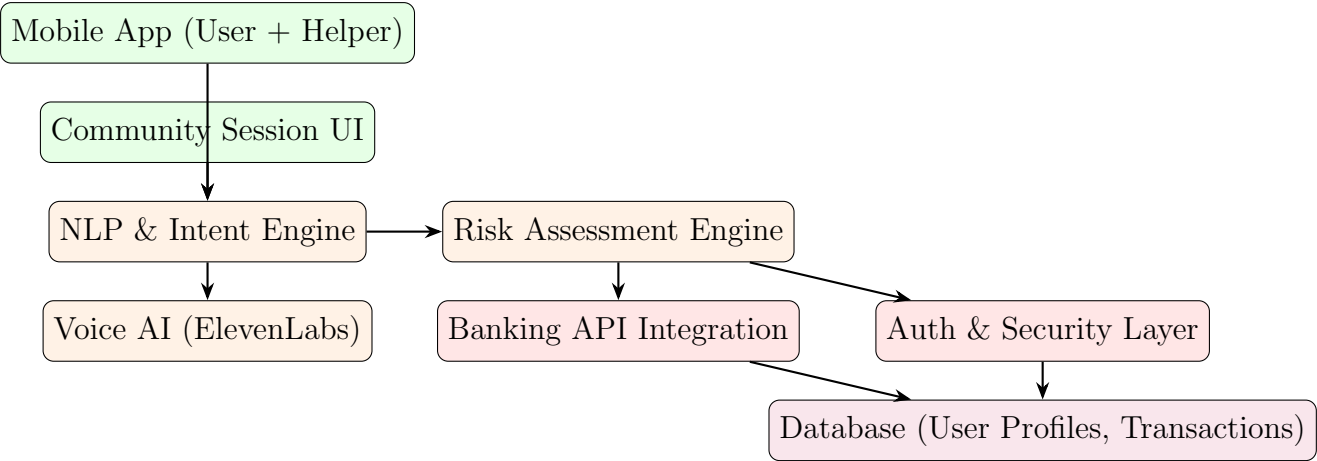


Figure 4: BankSathi Technical Architecture Workflow

Workflow Stage	Metric	Target	How We Track
Onboarding	Completion Rate	≥90%	User analytics
Transaction Initiation	Time to start	≤30 seconds	App timestamps
Intent Parsing	Accuracy	≥95%	NLP validation
Risk Assessment	Detection Time	≤2 seconds	System logs
Helper Response	Response Time	≤60 seconds	Session logs
Confirmation	User Error Rate	≤2%	Transaction logs
Transaction	Success Rate	≥98%	Banking API response
Follow-up	User Satisfaction	≥4.5/5	Post-transaction survey

Table 20: Performance Metrics Across Workflow

3. **Community Building:** Group sessions reduce isolation.

4. **Cognitive Support:** Transaction stories make banking meaningful.

5. **Measurable Impact:** Clear metrics demonstrate success.
- The product is technically feasible, socially impactful, financially sustainable, and scalable across geographies and services. It doesn’t just solve the problem—it transforms the entire experience of banking for vulnerable populations.

15 References

1. Reserve Bank of India. (2023). *Report on Digital Banking Accessibility in India.*

2. World Health Organization. (2022). *Global Report on Assistive Technology.*

3. NITI Aayog. (2023). *Digital Inclusion Index.*

4. HelpAge India. (2023). *State of Elderly in India.*

5. National Sample Survey Office. (2022). *Disability Statistics in India.*

6. International Telecommunication Union. (2023). *Accessibility Requirements for ICT Products*.
7. Ministry of Electronics & IT. (2022). *Guidelines for Accessible Digital Products*.
8. Reserve Bank of India. (2023). *Payment Systems Vision Document*.
9. United Nations. (2022). *Convention on the Rights of Persons with Disabilities*.
10. McKinsey & Company. (2023). *The Future of Digital Banking for Vulnerable Populations*.